SMART INDIA HACKATHON 2025

DevYatra

Problem Statement ID - SIH25165

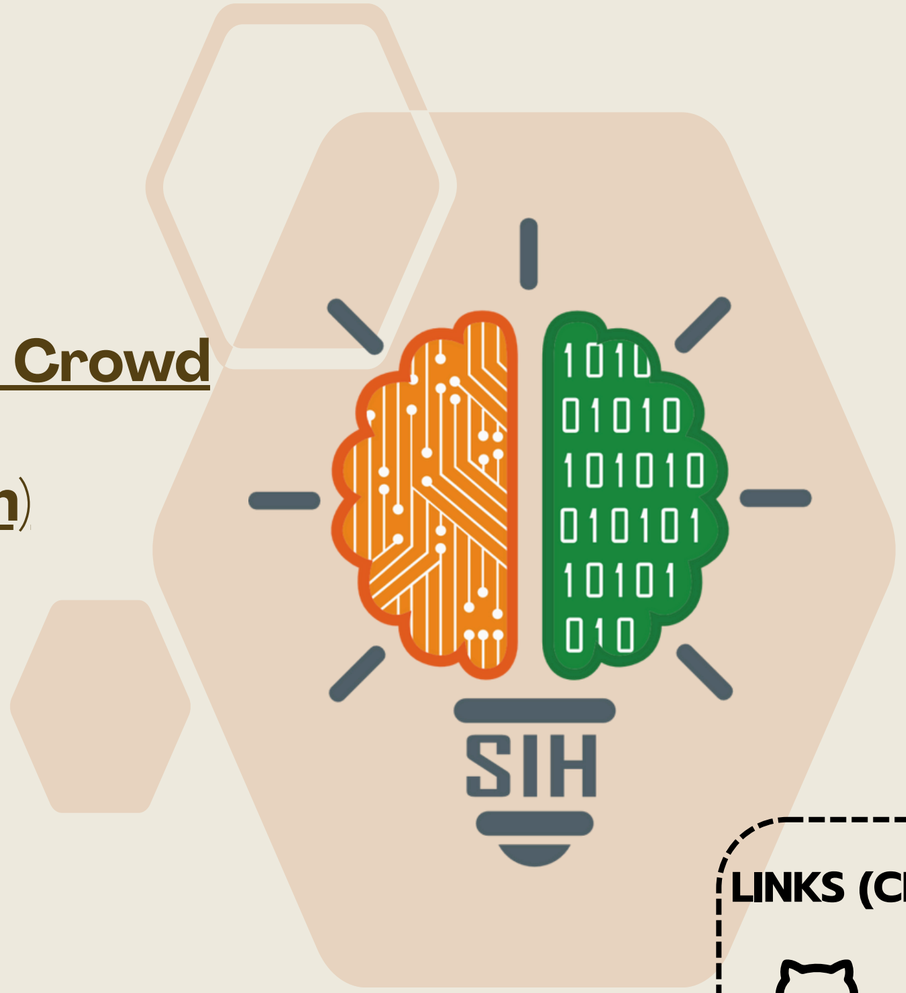
Problem Statement Title - Temple and Pilgrimage Crowd Management (Somnath, Dwarka, Ambaji, Pavagadh)

Theme - Heritage and Culture

PS Category - Software

Team ID - 65735

Team Name - Qbitron



LINKS (CLICK ME):

GitHub

YouTube

Qbitron

DevYatra

Guiding You Through Every Step

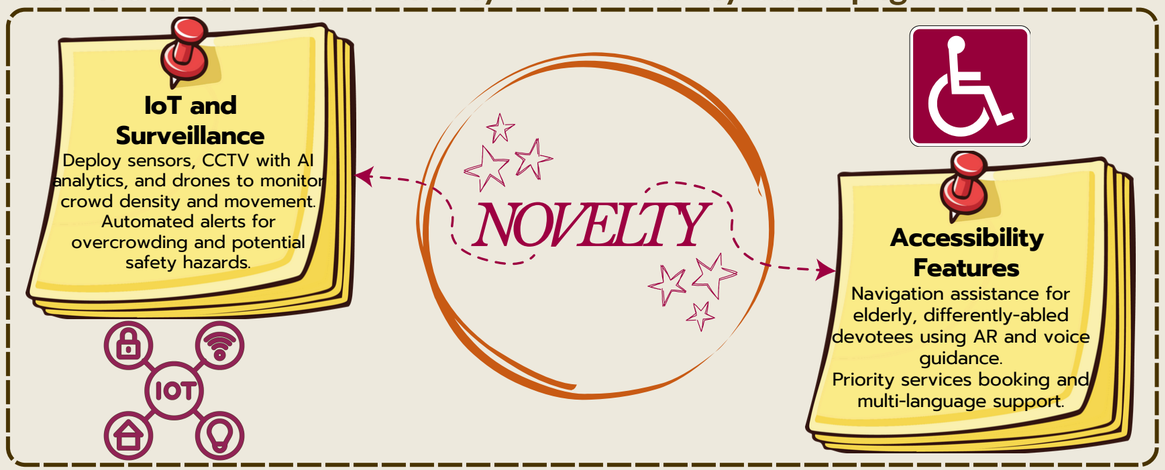
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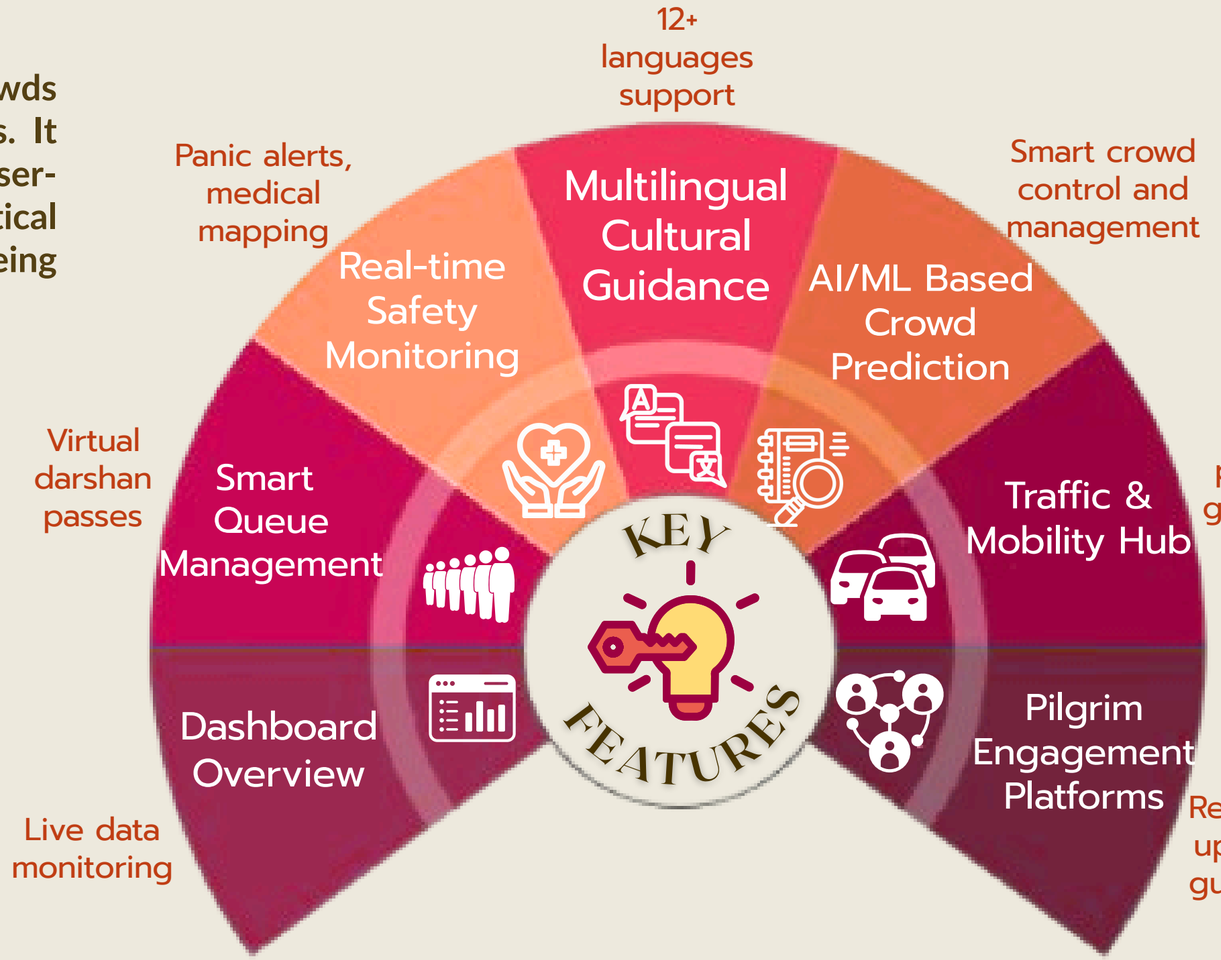
Overview

DevYatra is a unified digital platform developed to manage crowds efficiently and enrich the pilgrimage experience at major temple sites. It brings together streamlined entry processes, timely information, user-friendly guidance, and easy access to emergency help. Designed for practical usability, DevYatra aims to support both visitors and authorities, while being easy to implement and respectful of local traditions.

Problem Addressed

- Virtual queueing to cut long waits.
- Real-time monitoring for safety and emergencies.
- Live updates on routes, timings, and facilities.
- AI-based prediction for crowd and resource planning.
- Inclusive features for elderly and differently-abled pilgrims.





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TECHNICAL APPROACH

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FRONTEND

NEXT.JS

React 19

For server-side rendering and an interactive user interface.

- Web App: Shows live crowd data, maps, and temple updates.
- Admin Panel: Displays camera feeds, crowd density, and emergency controls.

BACKEND AND CORE

FLASK

For the backend and AI-heavy tasks.
Serves as the central brain of the system. It houses the core logic for all features, from crowd management to pilgrim guidance, making the system modular and scalable.

AI ENGINE

ultralytics YOLOv8

OpenCV

The platform's intelligence which work together.

- This is for object detection. Its specific function is to accurately detect and count every person in the live video feeds.
- Acts as the video processing layer. Its function is to capture and analyze camera feeds and then overlay the results for the admin dashboard.

DATA AND STORAGE

SQLite

A simple and efficient database is used for storing persistent data, such as user information, historical records, and safety reports. This allows for quick local access without the need for a complex external database.

PERSONALIZED AI CONTENT AND API'S

sarvam.ai

groq

Gemma

SerpApi

aviationstack

MapQuest

Geoapify

Radar

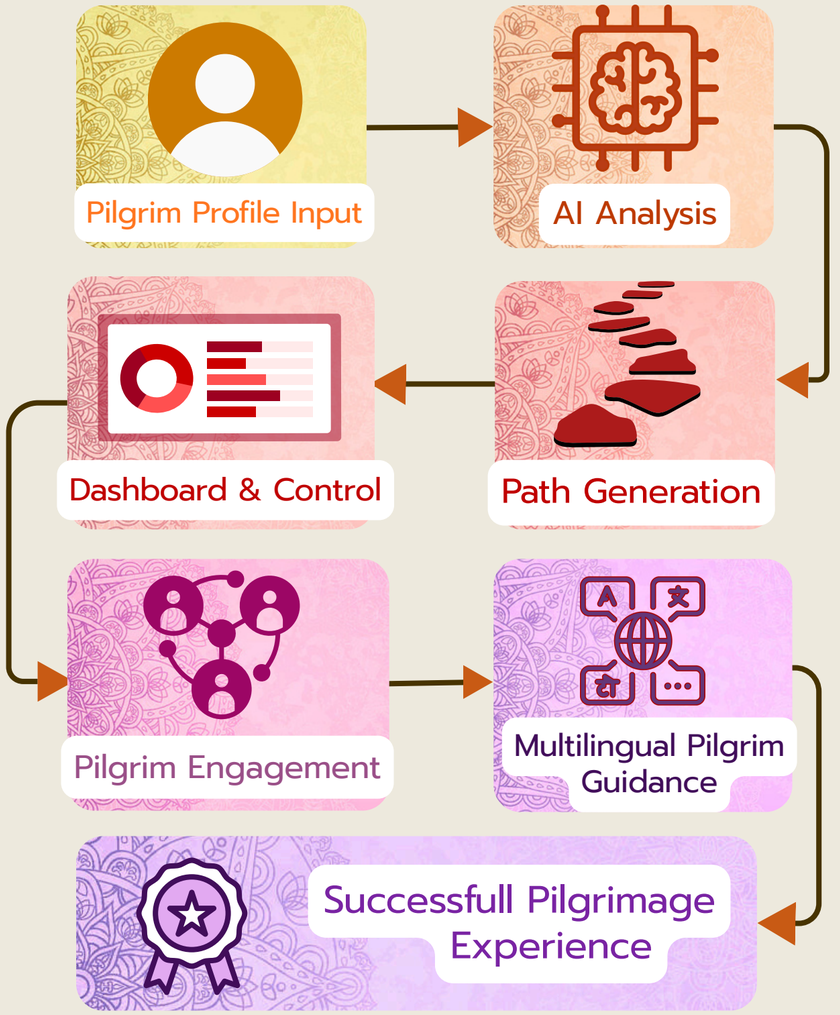
Pinata

OpenWeather

Leaflet

AMADEUS

PROCESS FLOW



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FEASIBILITY AND VIABILITY

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FEASIBILITY ANALYSIS

Market Demand
The pilgrimage sector in India is a huge and growing market.

Existing Infrastructure
Widespread smartphone and internet use provides a solid foundation.

Scalable & Secure
The platform's modular design and cloud architecture ensure scalability.

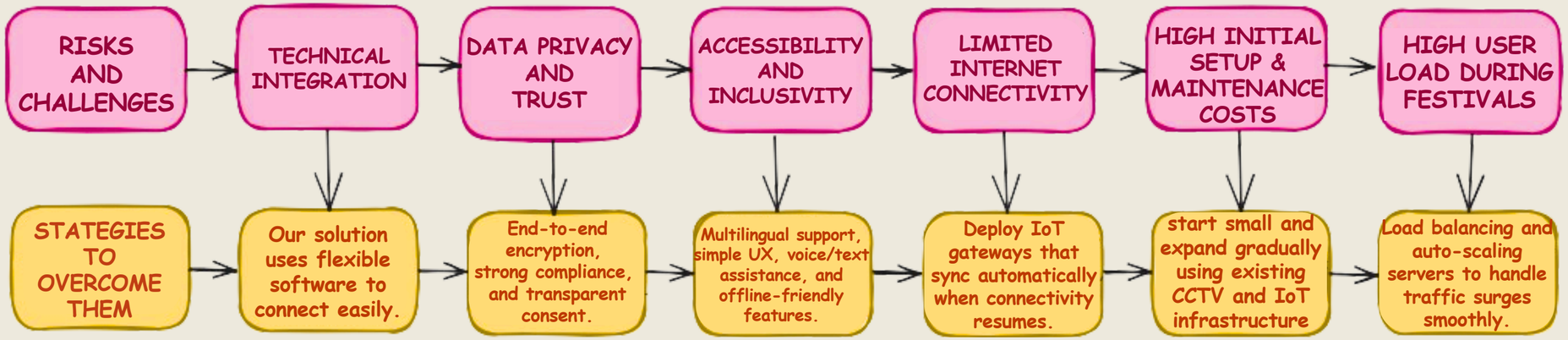
Government Alignment
The project aligns with government initiatives like the PRASHAD Scheme.

Strong Value
It addresses critical pain points like overcrowding and safety risks.

Viable Revenue
The project can generate revenue through subscriptions and service fees.

Industry Relevance
The solution is highly relevant to temple trusts and tourism boards.

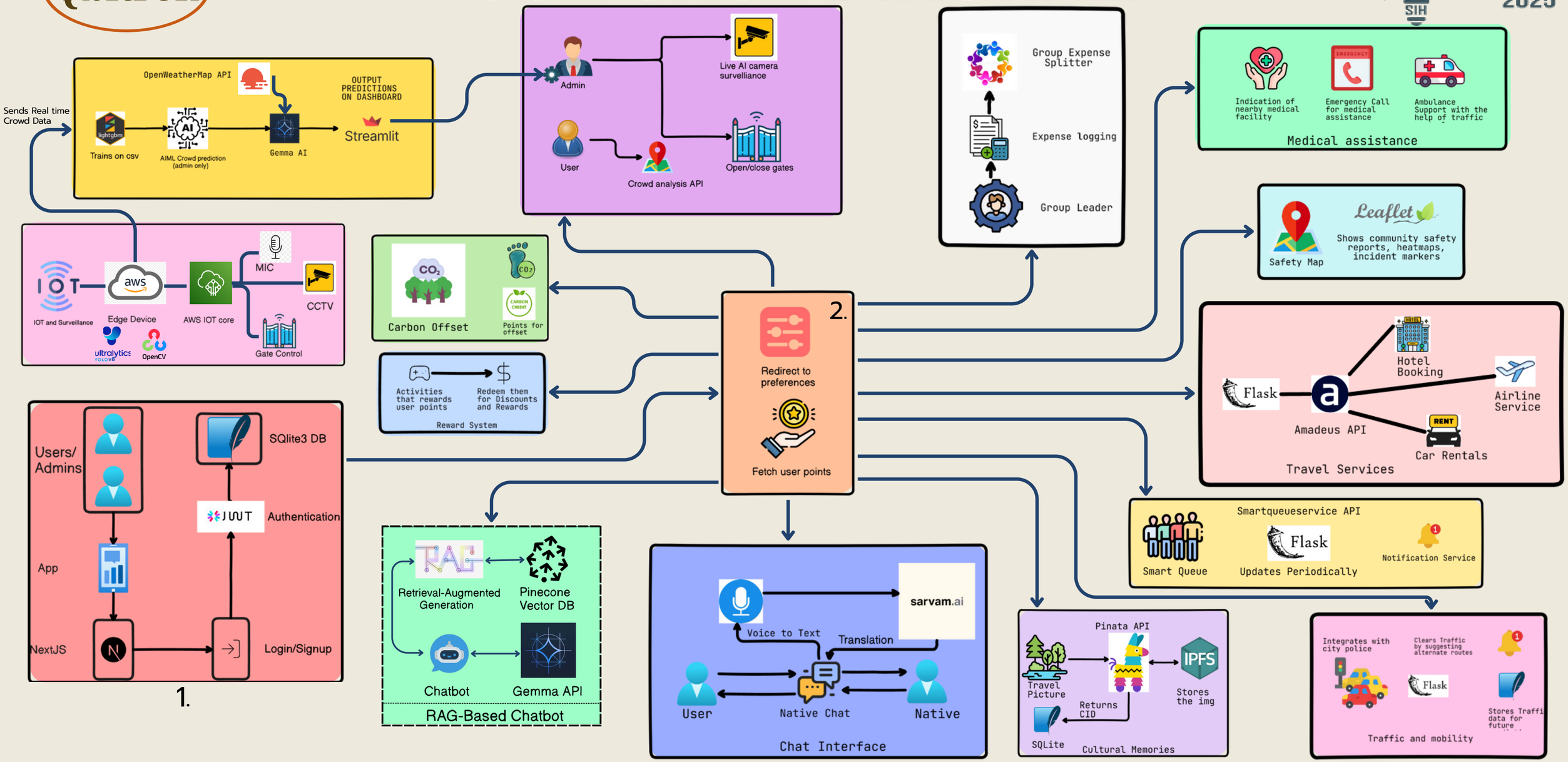
COMPETITORS	AI-POWERED CROWD PREDICTION	REAL-TIME SAFETY & EMERGENCY MONITORING	ACCESSIBILITY AND ASSISTED NAVIGATION	COMPREHENSIVE MULTILINGUAL GUIDANCE	END-TO-END PILGRIM EXPERIENCE
DevYatra	✓	✓	✓	✓	✓
TIRUMALA TIRUPATI DEVASTHANAMS	✓	✓	✗	✓	✗
SHREE JAGANNATH TEMPLE	✗	✗	✗	✗	✗
Sabarimala Virtual — A Kerala Police Initiative	✗	✗	✗	✓	✗



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USE CASE DIAGRAM

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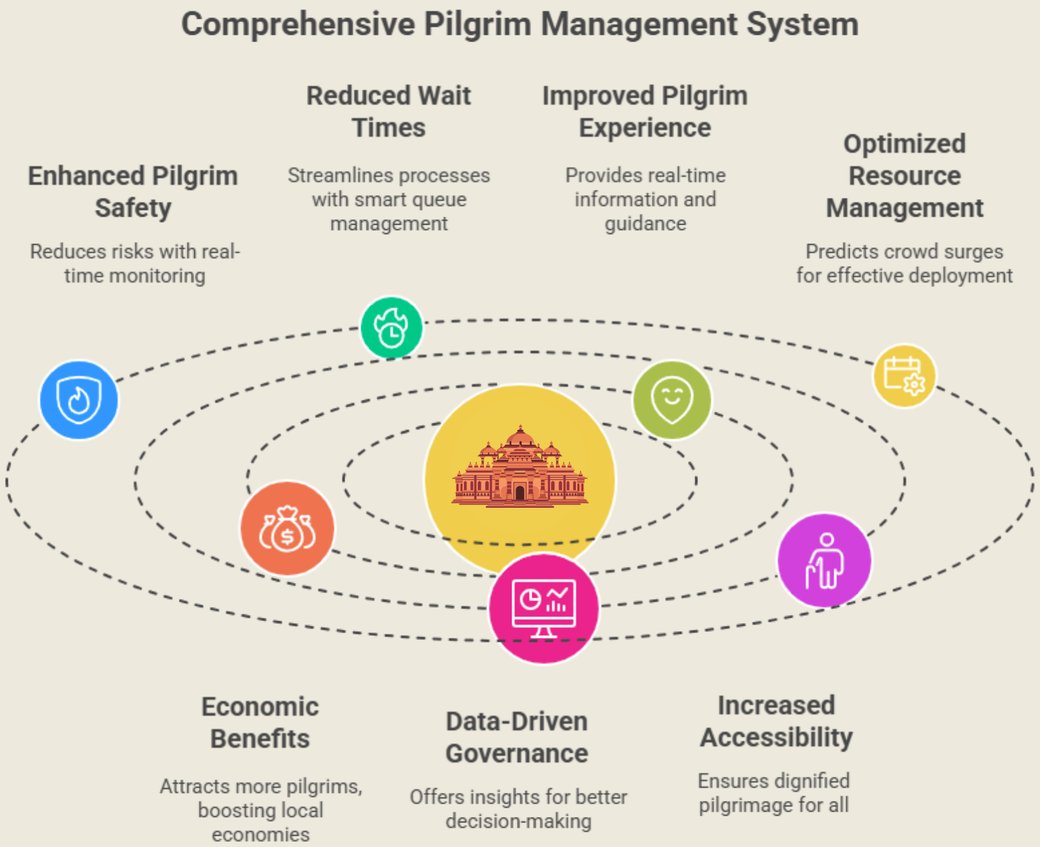


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IMPACT AND REFERENCES

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IMPACTS AND BENEFITS



RESEARCH AND REFERENCES

[1] N. Novera and Z. Ahmed, "Internet of Things (IoT) in smart tourism: A literature review," Spanish Journal of Marketing - ESIC, vol. 26, no. 3, pp. 325–339, 2022. [Online]. Available: https://www.emerald.com/sjme/article/26/3/325/351919/Internet-of-Things-IoT-in-smart-tourism-a

[2] A. Kontogianni, M. Alepis, and G. M. Giaglis, "Smart tourism: State of the art and literature review for the tourism domain," Smart Tourism, vol. 2, no. 1, pp. 1–12, 2020. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2590005620300059

[3] P. Suanpang and P. Pothipassa, "Integrating generative AI and IoT for sustainable smart tourism destinations," ResearchGate, Aug. 2024. [Online]. Available:https://www.researchgate.net/publication/383521863_Integrating_Generative_AI_and_IoT_for_Sustainable_Smart_Tourism_Destinations

[4] W. Wang, "Realizing the potential of Internet of Things for smart environments," IEEE Network, vol. 34, no. 5, pp. 192–199, Oct. 2020. [Online]. Available: https://dl.acm.org/doi/abs/10.1109/mnet.011.2000250

[5] J. Palomeque-Gonzalez, F. J. Ferrandiz-Colmeiro, and M. C. Gallego, "A modular, low-cost IoT system for environmental and behavioural monitoring in cultural heritage sites," arXiv preprint arXiv:2508.00849, Aug. 2025. [Online]. Available: https://arxiv.org/abs/2508.00849